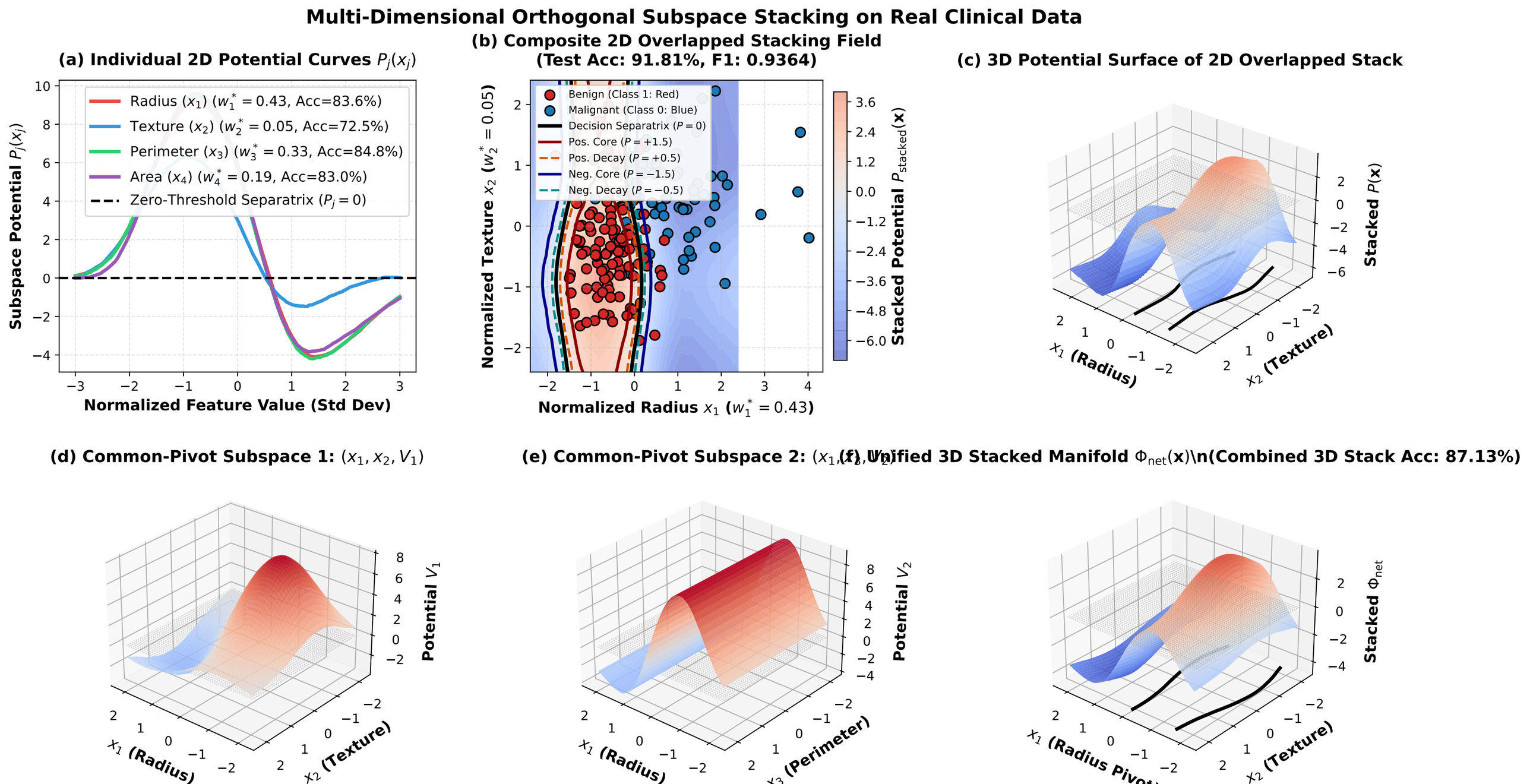

Multi-Dimensional Orthogonal Subspace Stacking on Real Clinical Data
(a) Individual 2D Potential Curves $P_j(x_j)$
(b) Composite 2D Overlapped Stacking Field
(Test Acc: 91.81%, F1: 0.9364)
(c) 3D Potential Surface of 2D Overlapped Stack
(d) Common-Pivot Subspace 1: (x_1, x_2, V_1)
(e) Common-Pivot Subspace 2: (x_1, x_3, V_2)
(f) Unified 3D Stacked Manifold $\Phi_{\text{net}}(\mathbf{x})$\n(Combined 3D Stack Acc: 87.13%)
Radius (x_1) ($w_1^* = 0.43$, Acc=83.6%)
Texture (x_2) ($w_2^* = 0.05$, Acc=72.5%)
Perimeter (x_3) ($w_3^* = 0.33$, Acc=84.8%)
Area (x_4) ($w_4^* = 0.19$, Acc=83.0%)
Zero-Threshold Separatrix ($P_j = 0$)
Subspace Potential $P_j(x_j)$
Normalized Feature Value (Std Dev)
Benign (Class 1: Red)
Malignant (Class 0: Blue)
Decision Separatrix ($P = 0$)
Pos. Core ($P = +1.5$)
Pos. Decay ($P = +0.5$)
Neg. Core ($P = -1.5$)
Neg. Decay ($P = -0.5$)
Normalized Texture x_2 ($w_2^* = 0.05$)
Normalized Radius x_1 ($w_1^* = 0.43$)
Stacked Potential $P_{\text{stacked}}(\mathbf{x})$
Stacked $P(\mathbf{x})$
x_1 (Radius)
x_2 (Texture)
Potential V_1
Potential V_2
x_3 (Perimeter)
Stacked Φ_{net}
x_1 (Radius Pivot)